

Recovery Modalities and When to Use Them

There are 168 hours in a week, and even at the high end of the training spectrum, it is rare for any endurance athlete to exercise more than thirty-five hours in a big week or more than twenty-five hours on an ongoing basis. Exercising—inclusive of running, strength training, yoga, Pilates, or other complementary activities like cycling, swimming, soccer, etc.—between twenty-five and thirty-five hours per week is 15 to 20 percent of the total time available in a week. It is far more common for ultrarunners to exercise for a total of ten to fifteen hours per week, some even fewer, which represents only 6 to 9 percent of the week. That means most ultrarunners, unless you have a particularly active job, are basically inactive 91 to 94 percent of the time.

When I lay this scenario out for some runners—particularly those with Type A personalities—their instinct is to view it as a challenge: “How can I increase the percentage of time I spend exercising?” They get standing desks, do lunges for no reason, and include a calf raise or two every time they encounter a set of stairs. But that’s not the challenge. As I will hopefully show you throughout this book, the time you spend running is rarely the limiting factor for performance, and just spending more time running is rarely the most effective or time-efficient way to improve performance. The challenge is how to use all that time you’re not running to maximize the quality and effectiveness of the time you are.

The Consequences of Doing Too Much

In order to make progress, stress and recovery have to work in harmony. That doesn’t always mean a perfect balance between the two. There will be times when your lifestyle, career, personal relationships, and training tip the balance toward more stress and less recovery. At other times, you may purposely reduce your sources of stress and tip the balance in favor of recovery. There are consequences to both directions. The simpler consequences come from too much rest and recovery; at first, performance will improve as physical and mental stress recede. However, when training frequency drops or time between training stimuli becomes too long, your workouts stop building upon one another. Your fitness and progress stagnate and can even start to slide backward.

Fortunately, studies examining the time course of detraining show that markers of fitness are quite resilient. While complete training cessation (meaning zero training) for several weeks can result in performance declines of >20 percent, adding in as few as three sessions (~50 percent training reduction) per week can keep those losses in the 5 to 10 percent range, even if the detraining phase is several weeks in length. In other words, if you cut your training in half, you are only going to pay a single-digit penalty for it. Furthermore, your return to your previous fitness will not take all that long (Bosquet, Laurent, and Mujika 2012; Lacour and Denis 1982; Mujika and Padilla 2000; Neuffer 1989; Pallarés et al. 2009).

At the other end of the spectrum, it can be more difficult to identify the consequences of tipping the balance toward too much work and not enough recovery. In order to create a training stimulus sufficient to induce a positive adaptation, you need to stress a physiological system to the point that performance starts to decline. The stress from an individual training session causes a small decrease in performance but is often not great enough to lead to significant long-term adaptations. You need a way to make training sessions additive. After individual training sessions, when you take adequate time for recovery, there’s a short-lived “supercompensation” effect that allows for increased performance. Consistent and proper training architecture schedules repeated bouts of exercise to take advantage of this cycle. Over a period of weeks, this cumulative stress leads to a decrease in performance known as functional overreaching, which is reversed by taking adequate time for recovery and is a normal part of the training process.

Pushing functional overreaching too far is what gets athletes in trouble. Instead of training to the point where performance diminishes and then letting up to allow for recovery, some athletes keep piling on the fatigue. Eventually, overreaching progresses to a scenario where more than two weeks of complete rest aren't enough to reverse extreme fatigue, disrupted sleep, irritability, and lack of desire to train. Some coaches and athletes refer to this scenario as “overtraining syndrome” (OTS), but training isn't the root of the problem—recovery is. The term “under-recovery” is more accurate, and the even newer term “unexplained underperformance syndrome” (UUPS) encourages athletes and coaches to look beyond the training plan to consider all sources of stress and rule out other possibilities, like RED-S, anemia, hypothyroidism, and acute or chronic illness (Lewis 2015).

Symptoms of OTS/UUPS include:

- Unrelenting feeling of fatigue
- Acute performance decline
- Sleep disturbances
- Elevated resting heart rate
- Low HRV
- Mood disturbances
- Loss of libido
- Low motivation to train
- Loss of appetite
- Frequent or persistent minor injuries and illnesses

The tipping point between functional overreaching and OTS/UUPS is highly individual and difficult to quantify, particularly because individual OTS/UUPS symptoms can be part of an athlete's normal response to hard and effective training. To the body, stress is stress, whether it comes from a workout, a hard day at the office, an argument with your partner, or an infection. The appeal of a multimodal approach to recovery is that it provides a wide range of tools to chip away at the monolith of stress.

The Goals of Recovery

Recovery is a broad term that encapsulates complete rest, restorative activities, and sometimes complex interventions, with the end goal of enhancing future performance in training and competition. It can be as simple and free as taking a nap, or it can be extremely complicated and expensive. It can also be very difficult to measure, and its effectiveness is often subjective and open to interpretation. This is why there are so many products, supplements, and modalities vying for athletes' attention and money; people want to believe these things work, the placebo effect is real, and it can be difficult to convince athletes to trust science over intuition and emotion.

To evaluate whether a recovery method is valuable, we need some criteria to test it against. We also have to recognize that recovery is very individual; what works for you may not work for someone else, and vice versa. There are a few products or methods that are utterly nonsense, but the vast majority provide varying levels of effectiveness for most athletes. I encourage athletes to test a potential recovery method against the following criteria:

- Will it promote muscle repair and growth, or any other adaptive response?
- Will it replenish something that training has depleted?
- Will it enhance your readiness for your next workout?
- Will it support immune system function?
- Will it be any more effective than getting more sleep?

That final question is perhaps the most important. As I'll explain shortly, sleep is the gold standard for recovery. For many athletes, adding activities in order to get more or better rest is counterproductive, and the best thing they can do is optimize their sleep behaviors.

Recovery Modalities

My goal here is not to go a mile deep into every possible method, supplement, and product purported to aid in recovery. For an in-depth treatise on the science of recovery, I recommend reading *Good to Go* by Christie Aschwanden. For the purposes of this book, I'm going to cover the recovery modalities my athletes use most and the ones athletes ask about most frequently.

TIME BETWEEN TRAINING

Proper scheduling of workouts and training phases is one of the most basic forms of recovery. This topic is covered extensively in Chapter 9 in the section on key workouts and in Chapters 10 and 12 in the sections on long-range and short-range plans, respectively. Nonetheless, it is important to mention here because a well-thought-out training strategy can be rendered ineffective without adequate recovery, and it will absolutely fail if it is both poorly structured and deficient in rest.

SLEEP

As I alluded to earlier, sleep is the gold standard and fulfills all the criteria for effective recovery. For starters, the physiological repair and growth that actually makes you better happens when you rest, and predominantly when you sleep. It improves motor learning (learning a new skill) and cognitive function. Additionally, better sleep can lead to better emotional well-being. Unfortunately, the converse is also true: a lack of sleep can lead to a decrease in optimism and mood.

None of this should come as a shock to athletes. You have experienced it firsthand when you woke up in the middle of the night to a crying child, barking dog, or unruly neighbors and felt slow and groggy the next day. You may have even snapped at your significant other for something trivial. It's OK, we've all been there. Because sleep has a major effect on training and recovery, it's important to understand how you can optimize your sleep to optimize performance.

Sleep Restriction Research

Many early studies into sleep restriction indicated that a few days of a few hours of sleep restriction had little to no impact on varying measures of strength and endurance performance. A 2015 review paper from Fullagar et al. noted that skill and precision sports are affected to a greater extent by sleep loss compared to endurance sports. The takeaway for athletes is if you have one or two terrible nights of sleep, you can probably just suck it up and get in a decent workout. You also have no excuse for a disastrous race due to one poor night's sleep.

As a coach, I rarely change an athlete's workout due to one night of poor sleep unless there is some other confounding factor like being on the verge of injury or illness, at the end of a long training block, or any other negative combination of objective and subjective measures. All this being said, it would be easy to look at this combination of research and practice and say, "Well, if it is not affecting performance, restricted sleep really does not matter."

Au contraire!

Effect of Sleep on Illness and Injury Risk

While missing out on a few hours of sleep for a few nights might not affect acute endurance performance much, the effect that high-quality sleep has on your immune system and your ability to recover (or not) is quite astounding. Both of these matter in the long run, as they affect week-to-week consistency, which is a far better indicator of success than any one workout.

A 2016 study by von Rosen et al. found that sleeping more than eight hours per night reduced the chance of injury by 61 percent in adolescent elite athletes. As a comparison, strength training, which has long been lauded for

reducing injury risk in endurance athletes, reduces the chance of injury by about the same amount, according to a 2018 meta analysis by Lauersen et al. The studies did not provide apples-to-apples comparisons, but the takeaway is that sleep has a profound effect on recovery and injury prevention and should be considered a vital component of any athlete's program.

Sleep affects your immune system similarly and just as dramatically. According to research from Cohen et al. (2009), adults who sleep fewer than seven hours per night for a prolonged period are nearly three times more likely to develop a cold. More recent review studies not only confirm the link between sleep disturbance and immunosuppression but also indicate that shorter sleep duration (less than six hours) may decrease the acute response to and lifetime protection from standard vaccines like influenza and hepatitis B (Irwin 2015; Walsh 2018). The effect of sleep on vaccine response gained even more importance in 2020 as the world pinned its collective hope on a vaccine to end the COVID-19 pandemic.

The simple takeaway is that sleep matters. It matters a little bit on the acute performance side of things, but more importantly, it matters more for long-term health, injury prevention, and training consistency.

The Architecture of Sleep

Your long-term habits around sleep make more of a difference than whether you slept well last night. As a result, I encourage athletes to think of “sleep hygiene” as habits to adapt to rather than a short-term issue to fix. In order to understand sleep hygiene, it is important to understand the basics of what goes on before, during, and after you sleep.

Figure 8.1 below illustrates how three critical areas—core body temperature and the hormones cortisol and melatonin—are affected just before, during, and after sleep. Cortisol is a hormone that prepares you to deal with stress, and melatonin is a hormone that helps regulate the sleep-wake cycle. Notice that core body temperature and cortisol go down and melatonin goes up right as sleep onset occurs. The key to unlocking powerful sleep lies in setting up your room and your habits to take advantage of the constellation of these changes.

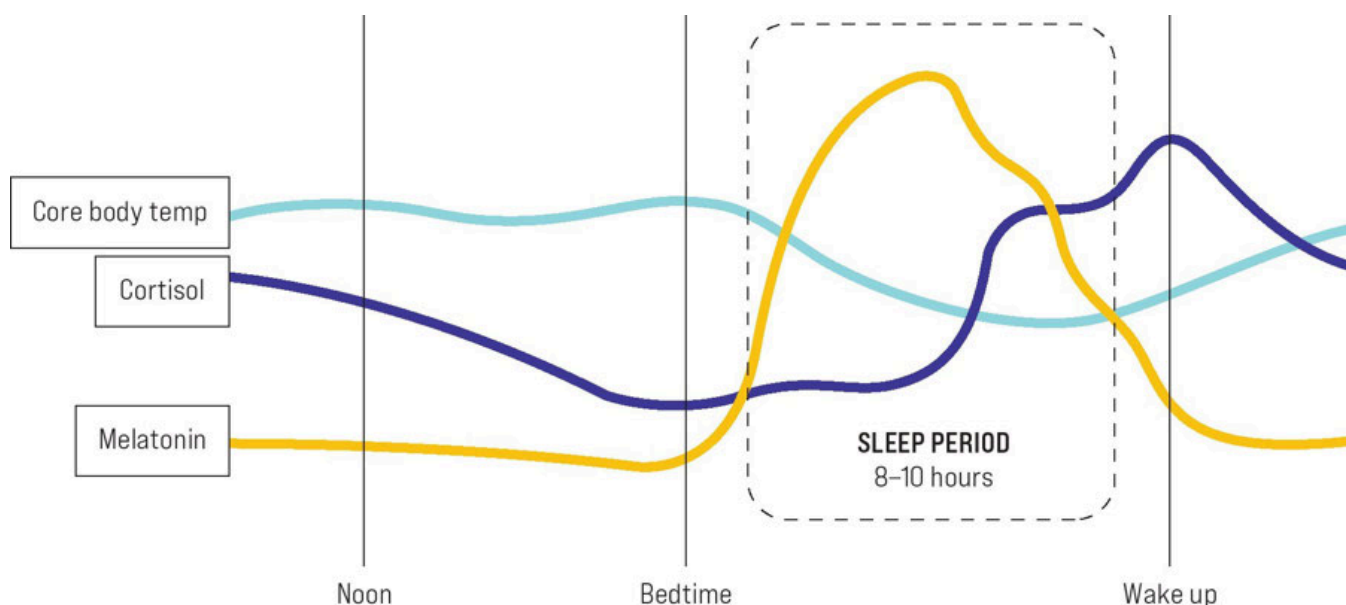


FIGURE 8.1 How core body temperature, cortisol, and melatonin fluctuate throughout the day.

Steps for Better Sleep Habits

STEP 1: SET UP YOUR ROOM

You should be able to describe your bedroom with the following words: calm, cool, dark, and quiet (Venter 2012). Optimizing these factors not only helps you get to sleep faster, it also helps increase sleep efficiency, which is your total sleep time divided by the time you're in bed. Sleep efficiency can have a big impact on immune function. According to a 2009 study on sleep habits and susceptibility to the common cold, "participants with less than 92% efficiency were 5.50 times more likely to develop a cold than those with 98% or more efficiency" (Cohen 2009).

- **CALM** : Minimize clutter and remove TVs, handheld screens, and bright alarm clocks.
- **COOL** : Ideal room temperatures are between 60°F and 67°F, which may feel cold but facilitate the natural drop in body temperature described above. Using blankets in a cooler room allows you to more easily establish and maintain a core body temperature conducive to sleep than a warm room and lighter sheets.
- **DARK AND QUIET** : Light (and the absence thereof) has the biggest effect on telling your body when it is time to sleep and time to be awake. Use blackout curtains to minimize light and sound seeping through the windows. Remove or cover objects with indicator or power lights.

STEP 2: ESTABLISH YOUR BEDTIME ROUTINE

A consistent sleep/wake time, combined with enough sleep (more than eight hours), will establish much of the routine. The rest is comprised of turning off any electronics—and your brain—ninety minutes before bedtime. Blue light emitted from your smartphone suppresses the release of melatonin (see above why melatonin release is important), and blue light filters on your phone and tablet only go so far. Bright lighting suppresses melatonin, too, so lower the lights or partially darken the room even before starting to get ready for bed. Create a routine where, instead of checking e-mail or Twitter, you journal, read for pleasure, meditate, or take a warm shower just before bed.

STEP 3: ESTABLISH YOUR WAKE-UP ROUTINE

Your wake-up routine is just as important. Regardless of workout schedule or traveling to a new time zone, try to wake up at the same time every day. What happens if you stay up late? Take a nap. This will maintain your bedtime and wake time routine. If you can't take a nap, then modify your normal wake-up time as best as you can so it can stay consistent.

STEP 4: ESTABLISH YOUR MORNING ROUTINE

Your routine after you rise (whether it's morning or another part of the day) is just as important as your routine before you sleep and can follow the subsequent steps: seek light, move, and nourish. As previously mentioned, light will have the biggest effect on signaling to your body that it's time to lay down or to rise. Therefore, seek light (preferably sunlight) when you rise. If you get up before sunrise, turn up the lights in the room where you get ready for the day.

After you are up and have some light, it's time to move. This can be your morning workout or preworkout routine or simply getting some chores done around the house. I recommend getting in ten to twenty minutes of movement first, before settling down to read the morning paper or heading out the door for your run. While you are getting in this first-of-the-day movement, or when it is done, nourish yourself with some fluids (because you will be dehydrated after sleeping) and your preworkout snack or breakfast.

ENERGY BALANCE

If sleep is the gold standard for recovery, energy balance is next in order of priority. Put simply, total energy intake is the linchpin for achieving adequate recovery and getting a positive response to training. Regardless of the macronutrient composition of what you eat, you will not significantly improve your running performance unless your total energy intake is high enough to support your basic bodily functions, your daily activity level, and the energy expenditure of training.

Athletes get into trouble with energy balance by undereating, overexercising, or a combination of both. Often, the mismatch between energy intake and energy expenditure stems from efforts to lose weight or maintain a low body weight, but the reason behind the mismatch can be more complicated than not eating enough after workouts. Athletes in sports that place a premium on being lightweight (see "The Body-Weight Conundrum" sidebar) can feel internal or external pressure to lose weight or look a certain way. This can be a factor in the development of behaviors and beliefs that can lead to negative self-image, disordered eating, overexercising, and a long list of physical and psychological problems. From a training perspective, RED-S (discussed in Chapter 2 in the "Considerations for Female Athletes" section, but applies to both males and females) poses the greatest risk to athletes who are acutely or chronically undereating, overexercising, or a combination of both.

RED-S was originally referred to as the Female Athlete Triad based on three key symptoms: amenorrhea, low energy availability, and decreased bone-mineral density. The change in terminology is more than words; it reflects an expanded understanding that energy deficiency is a problem that affects both males and females, and that it has far-ranging consequences. The host of performance complications illustrated in Figure 8.2 include decreased muscle strength, decreased bone mineral density, increased injury risk, and decreased training response, to name a few.

THE BODY-WEIGHT CONUNDRUM

BODY WEIGHT HAS AN IMPACT on the energy cost of running, but lighter isn't always better for performance, recovery, health, or emotional well-being. To understand the relationship between body weight and performance, let's take a look at what happens when you lose weight:

Energy Cost of Running: Normally on flat ground the energy expenditure of running is independent of speed. When you run faster, you burn more calories per minute, but you also cover distance more quickly, so the overall caloric expenditure per kilometer or mile stays relatively constant. When you lose weight, you reduce the energy cost of running by 1 percent for each 1 percent reduction in body weight. It might not seem like much, but over the course of 160K (100 miles), a 5 kg weight loss translates to an 800-calorie savings.

Ground Reaction Forces: From Chapter 5, remember that vertical GRFs are typically two-and-a-half to three times your body weight, so any reduction in body weight will lower GRFs.

Muscle Damage: We know a lot more muscle damage is incurred during an ultramarathon than during a marathon. To a point, athletes carrying more muscle may be able to spread the damage across more tissue and hence cope with the damage better than lighter athletes with less muscle mass. This may partly explain why top athletes in ultrarunning have more muscular builds than extremely lean marathoners.

VO2 max: The maximum amount of oxygen your body can take in, transport, and utilize per minute is often expressed relative to your weight. When we say a 70 kg elite ultrarunner has a VO2 max of 65 ml/kg/min, it means he can utilize 65 ml of oxygen for each kilogram of body weight in a minute. When you reduce body weight, absolute VO2 max stays pretty constant, but relative VO2 max goes up. Increasing VO2 max, whether through weight loss or intervals or both, impacts performance at all levels of intensity. Not only does being lighter make you faster at the very highest intensity you can achieve, but your aerobic pace and your lactate threshold pace get faster, the length of time you can sustain a given pace increases, and you recover from efforts above lactate threshold more quickly.

While the list above seems to favor being as light as possible, the bigger picture shows that proactively working to lose weight often does ultrarunners more harm than good. Carrying more muscle mass in the legs can correlate with faster ultramarathon finishing times (Knechtle, Rosemann, and Lepers 2010), although it is worth noting that training volume and finishing times at the marathon distance were better predictors of ultrarunning performance than any body measurement (Knechtle et al. 2009; Knechtle, Rosemann, and Lepers 2010; Knechtle and Senn 2011). This supports my preference to focus on cardiovascular development and the limiting factors for performance outlined in Chapter 4. By doing so, weight management tends to take care of itself as a result of increased training volume and intensity. Coaches who focus on weight loss as a means to improve performance risk creating abusive environments and almost always forego meaningful performance improvements that would result from better training.

Athletes don't need to have every symptom of RED-S to be negatively affected by undereating or trying to maintain an excessively light body weight. When you are losing weight because you are not sufficiently supporting your training workload with calories and nutrients, you will lose both fat and lean muscle mass. Because some of this muscle mass is important for coping with the metabolic stress of running long distances, losing weight by losing muscle is problematic. Athletes who are too lean cannot train as effectively because they cannot recover from hard efforts as quickly. Being too lean also puts additional stress on the immune system, leaving some athletes more susceptible to illness. When I encounter a runner who is not responding positively to training, is struggling to finish races or training sessions, and is constantly dealing with small colds, gastrointestinal bugs, and nagging injuries, the first thing I look at is the athlete's nutrition program. It is likely the athlete is not consuming enough calories to support his training workload.

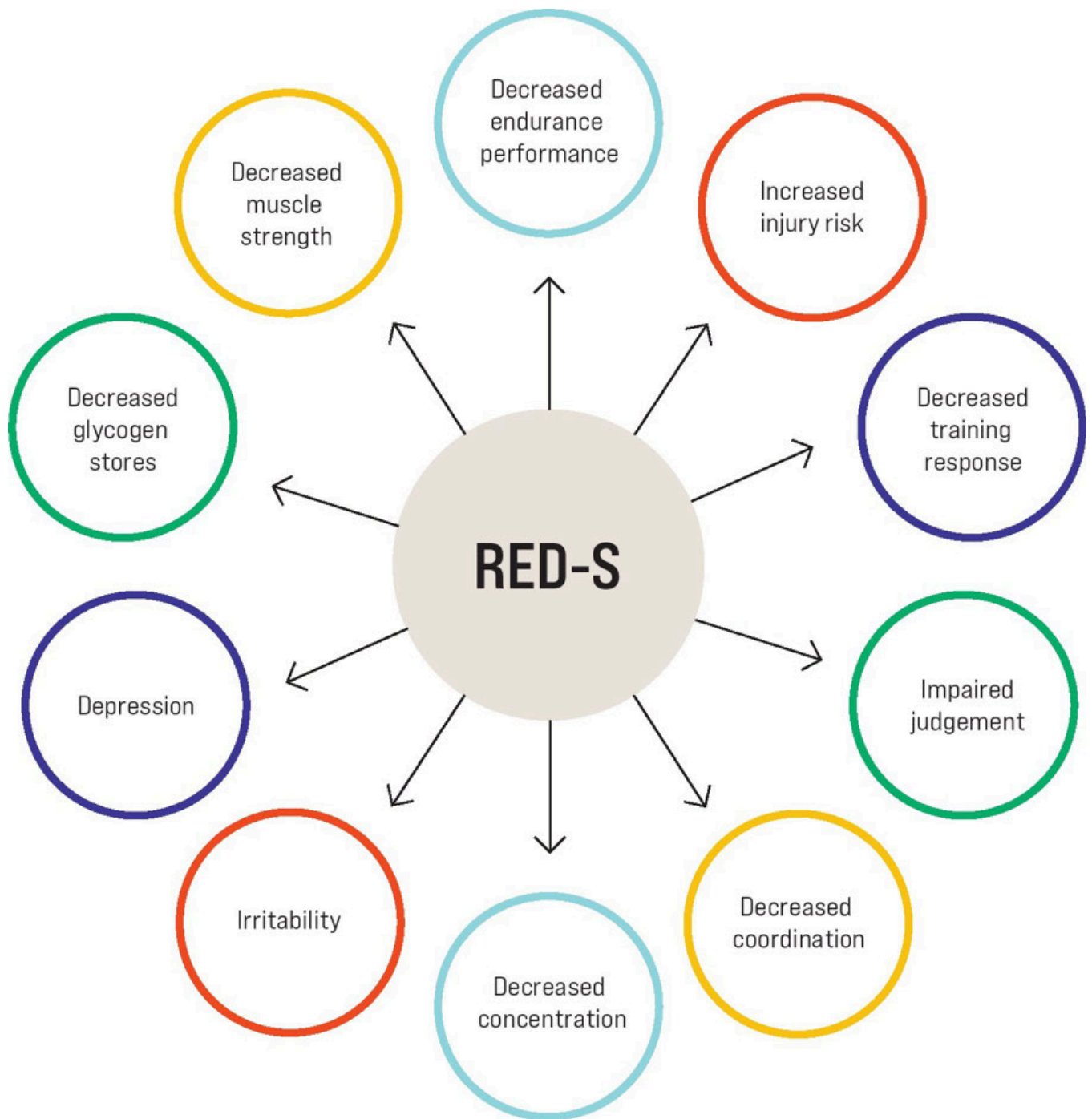


FIGURE 8.2 Effects of RED-S. Adapted from Mountjoy et al. 2018.

Postworkout Cooldown

Ultrarunners who have limited training time sometimes cut their warm-up or cooldown short or eliminate the cooldown altogether. From a recovery perspective, “active recovery” or reducing your pace to a light jog or walk for ten minutes at the end of your workout is time well spent, particularly after higher-intensity sessions. While you are exercising, skeletal muscle contractions assist the circulatory system to pump blood throughout the body. Blood has a lot of jobs, from delivering oxygen and fuel to muscles to dissipating heat, producing sweat (from plasma), redistributing lactate, and removing metabolic waste. During exercise, demand for all these functions increases, and that demand doesn’t return to baseline levels the moment you stop your GPS watch.

Active recovery at the end of a workout maintains somewhat elevated circulation, with the notion that this eases the body's transition back to baseline conditions. A 2018 review of active and passive recovery studies indeed showed that active recovery can accelerate recovery of lactate in blood and can bring core temperature down to normal more quickly (particularly if the activity maintains airflow over the skin). However, on balance, the review found that active recovery—including stretching—didn't improve actual performance in training or competitions the following day, compared to passive recovery. Put another way, active recovery delivered some benefits more quickly after exercise, but by the next day the recovery effects evened out (Van Hooren and Peake 2018).

While the physiological advantages of active recovery over passive recovery are more pronounced following high-intensity training than EnduranceRuns, I still incorporate active cooldowns at the end of athletes' workouts. They are perhaps most useful as a means to transition from the focus and intensity of a hard workout to the next parts and priorities of your day. I don't, however, prescribe a specific set of cooldown activities, because one isn't particularly more useful than any other. The main takeaway from the research is that the performance penalty for skipping or shortening your cooldown is likely minimal.

Compression/Massage/Foam Roller

I'm grouping these modalities together because they broadly do the same thing: apply pressure. Compression clothing and pneumatic compression devices aim to facilitate circulation of blood back to the heart and reduce swelling by moving lymph or preventing it from pooling in your extremities. Massage palpates muscle groups, sometimes superficially and sometimes more vigorously, often with the stated goals of pushing lactate and/or waste products out of the muscle, increasing blood flow to bring in nutrients, and addressing adhesions between layers of fascia (the connective tissue that surrounds muscles). The goals of foam rolling are virtually identical to massage, although the self-massage aspect may impart additional neural benefits on sensations of muscle soreness.

The research doesn't point to a clear physiological benefit from any of them, or at least no benefit that consistently improves performance. Compression garments and devices are absolutely effective for people following surgery, who are bedridden, or who have clinically poor circulation, but for healthy individuals the circulatory system doesn't really need extra assistance. Similarly, lactate clearance from muscles happens naturally, whether you rub the muscle or not (and the rate of lactate clearance is not important for recovery anyway). And while it is physiologically plausible for layers of fascia to stick to one another, the necessity or benefit of breaking those adhesions with massage or foam rolling hasn't been proven (Groner 2015; Wilke 2017; Zaleski 2016).

Compression, massage, and foam rolling make people feel good, enhance mind-body connection, and provide a relaxing time to pause. And there's a big placebo effect. These benefits are valuable on their own and meet the criteria of enhancing readiness for the next workout or competition by reducing stress. However, as with active recovery activities, the research indicates you won't be at a distinct disadvantage by skipping massage, compression, or foam rolling if you don't like them or they're not available.

THE LONG-HAUL FLIGHT HOME

THERE ARE INCREDIBLE ULTRARUNNING EVENTS all over the world, and as ultrarunners travel far and wide to participate, it is important to look at strategies for optimizing health, performance, and recovery on long-haul flights. For people with underlying health problems, sitting for long periods (especially more than four hours) increases the risk of deep vein thrombosis (DVT, characterized by a blood clot, often in the lower leg) and venous thromboembolism (VTE, when that blood clot travels through circulation and can end up in the lungs as a pulmonary embolism). Dehydration from the dry (1 percent humidity) cabin environment exacerbates the potential for DVT. These risks do increase for everyone somewhat, but for healthy populations and athletes, even the increased risk is extremely low and can be lowered still by staying hydrated and occasionally walking around the cabin (Bartholomew 2011).

Compression socks are recommended to help prevent VTE in people whose underlying health issues put them at higher risk for blood clots. Otherwise healthy endurance athletes can also be at increased risk for VTE, particularly after hard and/or long workouts or competitions, based on Virchow's triad: venous stasis, damage to veins, and hypercoagulability (Kushner 2020). The stasis comes from sitting still. The damage to veins can result from microtraumas during training and competition. And the hypercoagulability results from a disruption of hemostatic balance, which is the interaction of factors that promote clotting (coagulation) and factors that prevent it (fibrinolysis). Markers for both increase during and after exercise, a phenomenon known as hemostatic activation (Kupchak 2016; Zaleski 2015). Normally, markers go up and come back down, and you're none the wiser. The risk for VTE increases when your actions (sitting still for long periods, dehydration) and intrinsic factors (damaged veins, predisposition to clotting) tip the balance toward coagulation.

There's more research on the effect of compression socks on running performance when the socks are worn either while traveling to a race or during a race. There is little direct research on the effects of wearing compression socks on airplanes after a race. Findings are mixed, but there is some evidence that compression socks when worn during a race or hard workout could reduce subjective perceptions of delayed-onset muscle soreness and could attenuate hemostatic activation (thus reducing blood clot risk) in the hours after long and hard runs (Zaleski 2016). The only thing the research seems to consistently agree upon is that compression socks are safe for athletes before, during, and after hard exercise.

The upshot: on long-haul flight, or any long-duration travel after an ultramarathon, it is important to address the problems of Virchow's triad. Get up and walk at regular intervals. Stay hydrated. And if you want to use compression socks, they might help.

Cold Therapies

Before and during exercise in hot weather, ice baths, cold water immersion, ice vests, ice socks, ice slurry beverages, and cold wet towels are all effective ways to lower skin and core temperature and keep athletes from overheating. After exercise, these same options can be used to reduce body temperature if necessary, but assuming body temperature is normal or just slightly elevated from exertion, ice baths, immersion, and cryotherapy won't enhance recovery in a way that will improve performance tomorrow.

Exposure to cold was once thought to aid recovery by reducing inflammation. Ice baths, cold water immersion, and a few minutes in a chamber chilled to -140°F all rapidly lower skin temperature, leading to dramatic vasoconstriction in the extremities. There is evidence that cold exposure can lessen the inflammation caused by exercise, but it turns out that reducing inflammation is counterproductive in terms of long-term adaptation to training stress (Peake 2017). Your body's response to the inflammation caused by training is integral to making the fitness gains you're seeking, and blunting that response hinders adaptation.

Cold exposure also has an analgesic effect by numbing pain temporarily. For athletes performing repeated bouts of exercise with limited recovery, like track runners competing in preliminary and final heats, ice baths or other forms of cold exposure can provide temporary pain relief but may reduce force output in subsequent efforts. This is less of a concern for ultramarathon runners who rarely—if ever—perform multiple high-intensity track sessions in a single day.

Ice baths and contrast baths have such a long tradition in sports that I don't expect athletes will stop using them, and we will still see athletes getting into cold, picturesque mountain streams after runs. Like the other modalities discussed in this chapter, people like them because they believe they work and because they are part of the rituals of their sport.

Stress Relief

Goal-oriented athletes sometimes view recovery as the absence of physical activity, but there's a difference between sitting at peace and sitting still while stressed out or anxious. Psychological stress from the pressures of your life, career, relationships, and finances have to be taken into account when adding up the total stress that needs to be balanced through rest and recovery. This includes anxiety or runaway thoughts about your training, an upcoming workout, a previous workout, or the event you're preparing for. Feelings of stress can lead to frequent or continuous stimulation of the sympathetic nervous system (SNS)—the “fight or flight” one—which then leads to chronically elevated levels of hormones like cortisol.

Your parasympathetic nervous system (PSNS) regulates the “rest and digest” functions and unwinds the sympathetic nervous system's responses to stressful conditions. When athletes track HRV, the desired outcome is higher variability in the intervals between heartbeats. This is a sign that the PSNS is effectively modulating minute changes in excitability without disturbing heart rhythm. When HRV is low, it is a sign an athlete is not recovering well because the SNS is still overactive and causing minute fluctuations in the intervals between heartbeats.

Recovery modalities seek to re-establish homeostasis by activating the PSNS to dial back SNS activity. Remember that a drop in cortisol is part of the process of falling asleep, and the steps recommended for promoting sleep include a cool, dark, uncluttered, and quiet environment. These calming steps tune down the SNS and tune up the PSNS so you can get a good night of restorative sleep. The plausible mechanisms of action for active recovery, yoga, massage, compression, and cold therapies all include some level of PSNS activation as well.

Your brain is the ultimate modulator of long-term balance between the SNS and PSNS, so it is critical to address the brain's role in stress relief. Mental skills training, meditation, breathing exercises, counseling, spending time with a pet, or floating in a sensory deprivation tank can all be effective for dialing down the stress response, restoring balance, and hence improving your ability to recover from training and make positive adaptations. No two brains or sets of life experiences are the same, so the stress relief strategies that work best for specific athletes are also highly individual. The biggest takeaway is that non-exercise stress is inescapable, but when you are better prepared to cope with non-exercise stress, you create a greater capacity to recover from exercise stress, which allows you to increase training workload and achieve greater physiological adaptations.

Pills, Potions, and Supplements

Effective and safe (and legal) recovery rarely comes out of a bottle. Yet that doesn't stop companies and athletes from advocating the use of nonsteroidal anti-inflammatory (NSAID) medications, opioid and non-opioid analgesics, cannabidiol (CBD), and entire warehouses full of nutritional supplements. Here's how to approach these pills, potions, and supplements in terms of recovery.

NONSTEROIDAL ANTI-INFLAMMATORY (NSAID) MEDICATIONS

There is quite literally a stack of research on NSAIDs—which include naproxen (Aleve), ibuprofen (Motrin/Advil), acetylsalicylic acid (Aspirin), celecoxib (Celebrex), and others—and their effects on endurance athletes. The findings and sentiments can be summarized as follows:

- The use of NSAIDs increases the risk of hyponatremia (low blood sodium) in endurance athletes.

- The use of NSAIDs increases the risk of acute kidney injury in endurance athletes.
- The use of NSAIDs is prevalent amongst participants in triathlon, cycling, running, and ultrarunning events.
- Athletes generally do not know the risks of NSAID use.
- NSAIDs have not been shown to improve performance.
- There is very little evidence that demonstrates NSAIDs improve injury outcomes.
- NSAIDs can perpetuate existing musculoskeletal injuries, primarily through masking pain but also by affecting the biochemistry of the healing process.

I am not going to mince words here. The use of NSAIDs in endurance and ultra-endurance sports is dangerous. The research is unequivocal. Even seemingly benign use (i.e., the recommended dosage on the bottle), when compounded by the stress of the race or training, can exacerbate existing musculoskeletal injuries. Of greater consequence though: NSAID use increases the risk of hyponatremia and acute kidney injury, particularly in hot environments. Both can cause serious, if not fatal, medical complications.

Used outside of sports and following the recommended dosage on the bottle or the instructions of your doctor, NSAIDs can alleviate occasional pain, inflammation, or fever with only low to moderate risks for stomach bleeding or kidney injury.

ANALGESICS

Pain is part of sports, and ultra-endurance athletes often pride themselves on how much pain they can tolerate. It is important to recognize that temporary soreness is part of the normal process of training and competing, and an accurate assessment of pain is essential for understanding how well or quickly you are recovering and hence how much or how soon you can go again. That doesn't mean you have to be a masochist and reject all manner of pain relievers, but it does mean that masking pain with strong analgesic medications can encourage athletes to accumulate an inappropriate amount of training stress or cause them to exacerbate a minor injury until it turns into a major one.

Acetaminophen is a common over-the-counter analgesic that is neither an NSAID nor an opioid. It is a pain reliever and fever reducer, but not an anti-inflammatory. Like NSAIDs, the occasional use of acetaminophen for pain relief or fever reduction following recommended dosages or your doctor's instructions carries few risks for negative side effects. The same cannot be said for the other major category of pain relievers: opioids.

Opioid and synthetic opioid medications, including morphine, fentanyl, tramadol, and oxycodone, are powerful analgesics that can play an important role in the acute and chronic management of severe pain. Sadly, athletes have been suspected of using opioids, particularly tramadol, as performance-enhancing drugs to mask the pain of competition. While tramadol is not on the World Anti-Doping Agency's List of Prohibited Substances as of 2020, it is in their monitoring program, and individual sports—including cycling—have banned its use in competition.

The exponentially greater problem with opioids is that they carry a significant risk for addiction and fatal overdoses. According to the US Centers for Disease Control and Prevention, in 2018 opioids contributed to the deaths of more than 46,000 Americans. Therefore, put simply: if an athlete is experiencing pain so intense it can only be managed with opioid analgesics, they are not healthy enough to continue training. An exception could be made for an athlete using opioids, under a doctor's supervision, in order to participate in physical therapy or post-injury rehabilitation.

CANNABIDIOL (CBD)

CBD is everywhere these days, in everything from tinctures and skin cream to food and shampoo. It is a phytocannabinoid found naturally in Cannabis plants, including Cannabis sativa (hemp) and Cannabis indica (marijuana). Unlike tetrahydrocannabinol (THC), which is also found in cannabis, CBD is not psychoactive.

Cannabinoids exist naturally in your body, and the endocannabinoid system is thought to modulate the activity of neurons by selectively inhibiting the release of certain neurotransmitters. For instance, when CBD is used to treat epilepsy, it may reduce seizure activity by, in part, reducing the buildup of glutamate, an excitatory neurotransmitter (Grewer et al. 2009). Proponents of CBD contend that consuming it could be thought of as supplementing or increasing the activity of your body's existing endocannabinoid system.

CBD is primarily marketed to athletes as a pain reliever, anti-inflammatory, sleep aid, and anti-anxiety aid—or, taken collectively, as a recovery aid. The research into the validity of these claims is still evolving. As a result, I approach CBD with a healthy dose of skepticism. It has bioplausible mechanisms of action that could prove useful, but there isn't enough evidence yet that it actually delivers meaningful benefits for athletes. That said, research shows that the effectiveness of several of the recovery modalities mentioned in this chapter is debatable, so as with those, if you believe CBD works and it makes you feel better, that may be enough for it to have a positive effect on recovery.

Effectiveness aside, there are bigger reasons I encourage caution when it comes to CBD:

- **YOU DON'T KNOW WHAT YOU'RE GETTING:** A 2017 study published in the Journal of the American Medical Association tested eighty-four different CBD products from thirty-one companies. They found that 21.4 percent of the CBD products tested also contained appreciable amounts of THC and only 31 percent contained accurate amounts of CBD compared to the label (Bonn-Miller et al. 2017). Ideally, market competition should gradually solve this problem, but one look at the ongoing inconsistency of ingredients and labeling in the much more mature nutritional supplement market suggests mislabeling will be a persistent issue. And the risk is real. Even though the World Anti-Doping Agency has increased the legal limit of THC for athletes in an effort to lower the risk of positive tests due to accidental or recreational use, exceeding that limit—even accidentally—will still result in a ban.

CBD producers can ask a third-party lab to perform an analysis on their products and produce a certificate of analysis (COA). Companies that have taken this step will typically have these COAs listed on their website. Look for companies who have taken this extra step to ensure that their products have what they say they have in them.

- **YOU DON'T KNOW WHO YOU'RE GETTING IT FROM:** Choose a CBD product manufacturer who can tell you where their CBD is coming from. States where recreational and medical marijuana is legal (like Colorado, California, and Oregon) have higher regulatory standards than other states. Currently there's a lot of clutter in the CBD marketplace. There are (relatively) established players with good processes and manufacturing procedures that could produce an efficacious product, but unfortunately there are also, shall we say, seedy players in the marketplace.

- **IT'S BEING MARKETED AS A PANACEA:** This should be a red flag, whether you're considering CBD or any other supplement. When you can't really prove what a product can do, it is hard to disprove what it can't do. Hopefully the industry will become more regulated, production will become more standardized, and research will home in on a reasonable set of effective use cases. We're just not there yet.

NUTRITIONAL SUPPLEMENTS

There's a lot of money to be made selling busy people drinks, pills, and powders that are supposed to replace what's missing from their diets or provide enhanced benefits they can't get from food. There are absolutely cases where supplementation is necessary to correct a clinical nutrient deficiency, but the consensus of reputable federations and sports science organizations is that few supplements provide performance-enhancing benefits that can be supported by solid science (International Olympic Committee Consensus Statement 2018). In 2016, the Academy of Nutrition and Dietetics, the American College of Sports Medicine, and the Dietitians of Canada released a position paper that stated, "Relatively few supplements that claim ergogenic benefits are supported by sound evidence" (Thomas, Erdman, and Burke 2016).

The Australian Institute of Sport (AIS) developed a system to categorize supplements based on their safety, their legality in sport, and the strength of the scientific evidence supporting their efficacy. The 2019 AIS Sports Supplement Framework features an ABCD classification system. Foods and supplements in the A category are "supported for use in specific situations in sport using evidence-based protocols." Category B foods and supplements are "deserving of further research and could be considered for provision to athletes under a research protocol or case-managed monitoring situation." In other words, they might work, but research hasn't proven they do work. Category C foods and supplements "have little meaningful proof of beneficial effects." The vast majority of supplements fall into this category. The AIS summarizes this by saying, "If you can't find an ingredient/product in Groups A, B or D, it probably deserves to be here." Category D, by the way, is reserved for performance-enhancing drugs and supplements that are "banned or at high risk of contamination with substances that could lead to a positive drug test."

GROUP A

OVERVIEW OF CATEGORY	SUBCATEGORIES	EXAMPLES
Evidence level: Supported for use in specific situations in sport using evidence-based protocols	Sports foods Specialized products used to provide a convenient source of nutrients when it is impractical to consume everyday foods	Sports drink Sports gel Sports confectionery Sports bar Electrolyte supplement Isolated protein supplement Mixed macronutrient supplement (bar, powder, liquid meal)
	Medical supplements Supplements used to prevent or treat clinical issues including diagnosed nutrient deficiencies. Best used with advice from an appropriate medical/nutrition practitioner	Iron supplement Calcium supplement Multivitamin supplement Vitamin D supplement Probiotics
	Performance supplements Supplements/ingredients that can support or achieve an enhancement of sports performance Best used with an individualized and event-specific protocol, with the advice of appropriate sports science/nutrition practitioner	Caffeine B-alanine Bicarbonate Beetroot juice/nitrate Creatine Glycerol

TABLE 8.1 Various supplements and evidence. Adapted from "Supplements." Sport Australia. www.ais.gov.au/nutrition/supplements accessed September 2020.

GROUP B

OVERVIEW OF CATEGORY	SUBCATEGORIES	EXAMPLES
Evidence Level: Deserving of further research and could be considered for provision to athletes under a research protocol or case-managed monitoring situation	Food polyphenols Food compounds which may have bioactivity including antioxidant and anti-inflammatory properties May be consumed in food forms or as isolated chemicals	Cherries, berries, and black currants Quercetin, ECGC, epicatechins, and others
	Other Compounds which attract interest for potential benefits to body metabolism and function	Collagen support products Carnitine HMB Ketone supplements Fish oils Phosphate Curcumin
	Sick pack Multi-supplement approach to address an issue of health or well-being Best used with advice from an appropriate medical/nutrition practitioner	Zinc lozenges and Vitamin C
	Amino acids Constituents of protein which may have effects when taken in isolation, or may be consumed individually by the athlete to fortify an existing food/supplement that is lacking in this amino acid	BCAA/Leucine Tyrosine
	Antioxidants Compounds often found in foods which protect against oxidation or reactions with free-radical chemicals May be consumed in food forms or as isolated chemicals	Vitamin C and E N-acetyl cysteine

The supplements in Category A and Category B can be found in Table 8.1. I didn't include Category C or D because they can be summarized as "everything else" and "performance-enhancing drugs and prohibited substances," respectively.

From a recovery standpoint, my biggest concern with supplements is that they can give athletes a false sense of confidence that makes them less vigilant about things that we know work. For instance, if taking a supplement makes you think you don't have to be as conscientious about making good food choices, the negative consequences of poor food choices are going to outweigh any benefits from the supplement. Replace "food choices" with "sleep" or "fundamental training" and the statement remains just as true. The effects of supplement use go beyond the efficacy of the substance you're consuming; they change an athlete's perception of what is valuable and impactful for improving performance. Some of my athletes consume supplements, but only after we have thoroughly investigated the physiological necessity, the athlete's motivations, and the anticipated outcome.

Nutrient Timing

Athletes inherently know that workouts burn energy and that eating is required to replenish that energy and provide fuel for adaptation. The logical extension is that coordinating the timing of your calorie and/or macronutrient intake with your training must be beneficial. There are instances where this is true, but overall, the importance of nutrient timing has been oversold.

Remember the research that indicated that active recovery provided benefits more quickly following exercise, but that with or without it athletes were equally recovered and ready for training the following day? The same is essentially true for the effect of nutrient timing on recovery. As I'll discuss more in Chapter 13, consuming carbohydrate and protein within thirty to sixty minutes after exercise (long referred to as the "anabolic window") can increase the initial rate of muscle glycogen replenishment, but the "window" doesn't slam shut after sixty minutes. Assuming an athlete's daily diet provides adequate total energy and a variety of macronutrients, muscle glycogen levels will be completely restored by the following day.

Similarly, there is research all over the map that claims to show performance, metabolism, recovery, or weight-loss benefits to eating breakfast, skipping breakfast, eating most of your calories for dinner, eating fewer calories late in the day, eating small meals more frequently, eating bigger meals less frequently, or restricting feeding hours to create fasting conditions. Some of the research is well done, and some of it has flaws in methodology or small sample sizes, but taken collectively the reasonable conclusion is that none of these nutrient timing strategies promotes enhanced performance or recovery for everyone all the time.

When it comes to the effect of nutrient timing on performance and recovery, here's what I tell my athletes:

1. Eating enough total energy each day is more important than how you distribute your energy intake.
2. When possible, protein consumption should be spread across the day rather than concentrated in one or two big feedings. You utilize protein around the clock for recovery, physiological adaptation, and immune function, but unlike carbohydrate and fat, protein can't be easily stored in the body for later use.
3. Rapid postworkout carbohydrate replenishment (leveraging the "glycogen window") is more important if you have fewer than eighteen to twenty-four hours before your next training session or competition. Otherwise, you can eat as you normally would, and the results will be the same.
4. Specific carbohydrate-availability strategies—either in the direction of carbohydrate loading or exercising with low carbohydrate availability—can play a role in training or competition but should not take priority over fundamental, high-quality, fully fueled training.

5. The eating pattern that works best for your lifestyle, personal preferences, and performance really is the one you should stick with. Be open-minded and try different patterns to see how they make you feel, but trust that what feels right probably is right.

KEY POINTS FOR RECOVERY

More activities aren't necessarily better for recovery than simply doing less between training sessions. Don't mistake being busy with being effective.

The two best things you can do to optimize recovery are getting more sleep and making sure your total daily energy intake is high enough to support your total daily expenditure.

Active recovery, nutrient timing, massage and compression, and/or cold therapies can make you feel better sooner, but within twenty-four hours the recovery benefits are roughly equal to passive rest, sleep, and adequate caloric intake.

NSAID pain relievers are dangerous for endurance athletes. Opioids are effective analgesics but have a high risk for abuse, addiction, and overdose.

Despite physiological plausibility, there isn't enough high-quality research yet to say whether CBD is an effective and consistent recovery aid. Nevertheless, it is here to stay.

The vast majority of nutritional supplements have little to no solid science to back up claims of improving performance, recovery, or health



